

Overview of the Ontology

This ontology is designed to model fundamental aspects of sandwiches, drawing upon the description provided in the Simple English Wikipedia article. The primary focus is to represent the structural and compositional elements that define different types of sandwiches. The ontology formalizes relationships between sandwich types, the ingredients they contain, and their structural configuration. It captures general categories such as open sandwiches, club sandwiches, and double-decker sandwiches, while also representing the core components like bread and fillings. Special attention is given to structural requirements, such as the number of bread layers, and to organizing ingredients into logical subtypes. Logical roles are employed to express stacking and type relationships within a sandwich.

The ontology comprises fifteen classes. These include Sandwich as the overarching concept, and eleven subclasses that represent specific sandwich types and components: Bread, Filling, OpenSandwich, DoubleDeckerSandwich, ClubSandwich, BLT, Panini, and ToastedSandwich. These classes collectively capture a range of sandwich variations.

In addition to the core classes, the ontology includes five object properties that define relationships between sandwich elements. These are hasIngredient, hasBread, hasFilling, isLayeredWith, and isTypeOf. The hasIngredient property serves as a general link between a sandwich and any of its components. The properties hasBread and hasFilling are introduced as sub-properties of hasIngredient. This structure allows the ontology to express both general and detailed statements about sandwich composition while maintaining logical consistency.

The class Bread is further divided into WhiteBread and WholeGrainBread, providing a means of distinguishing sandwiches based on the type of bread used. Similarly, the class Filling is subdivided into four categories: Cheese, Meat, Egg, and Vegetable. These subclasses reflect common types of sandwich fillings as mentioned in the source material. This hierarchical organization supports reasoning about sandwich content and enables detailed categorization or filtering based on dietary or ingredient-based queries.

The class Sandwich serves as the parent for all sandwich variants. OpenSandwich, DoubleDeckerSandwich, and ClubSandwich are all defined as subclasses of Sandwich, enabling structured inheritance of sandwich properties. The BLT is further modeled as a subclass of ClubSandwich. This structure reflects the real-world hierarchy of sandwich types and supports inference mechanisms within the ontology. The organization of object properties also contributes to this logical framework. hasBread and hasFilling are defined as sub-properties of hasIngredient, which allows for general rules to apply across all ingredients when needed, while maintaining the capacity for more specific assertions.

The property isLayeredWith is marked as transitive to represent the concept of stacking within a sandwich. This allows the ontology to model the idea that if one ingredient is layered with another, and that second ingredient is in turn layered with a third, then the first is indirectly layered with the third. This is particularly useful for modeling the internal arrangement of double or triple-decker sandwiches. The property isTypeOf is declared functional to ensure that an individual sandwich instance corresponds to exactly one

sandwich type. For example, a BLT can be defined as a specific individual that isTypeOf ClubSandwich.

A central axiom implemented in the ontology is the constraint that every sandwich must have at least two slices of bread. This is expressed in description logic as: Sandwich is a subclass of things that have at least two hasBread relationships with Bread, or more formally, $\text{Sandwich} \sqsubseteq \geq 2 \text{ hasBread.Bread}$. In natural language, this means that every sandwich must have two or more slices of bread. This axiom directly corresponds to a statement in the source text: "A sandwich is a food prepared by placing different kinds of foods between two pieces of bread." By enforcing this requirement, the ontology formalizes the structural minimum needed to qualify as a sandwich.

There are several advantages to building a sandwich ontology using this structure. It enables logical formalization of sandwich types and components, allowing for automated classification and structured queries. The hierarchical organization allows for straightforward extension to include cultural or dietary variations. Defining subclasses for ingredients and sandwich types increases semantic clarity and precision. The use of logical axioms ensures that only structurally valid sandwiches are represented.

However, this ontology also has limitations. It reflects a largely Western conception of sandwiches and may not easily accommodate regional or non-standard forms such as wraps or ice cream sandwiches. These edge cases might violate structural axioms and require exceptions or redefinition of certain rules. Modeling every possible filling as a separate subclass could lead to unnecessary complexity. Lastly, expressing certain real-world knowledge, such as a club sandwich requiring bacon, chicken, and salad, would require more detailed axioms or the use of data properties, which are outside the current scope.